

Single cell expression map of pancreatic islets using data from HPAP, IIDP and Prodo (V3 Freeze)

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Result summary

1. Release notes (major updates since V1 freeze)
 - a. Reprocessed raw data so that runs of the same library were merged prior to alignment, enabling correct handling of PCR duplication.
 - b. Added hashtagged-oligo samples.
 - c. Relaxed thresholds for number of cells per sample, enabling more samples retained.
 - d. Added an additional round of ambient RNA correction where model parameters were optimized for every sample.
 - e. Multiple doublet detection methods for better separation of cell populations
 - f. Code availability: All code for the analyses are provided at <https://github.com/PanKbase/snRNAseq-NextFlow> and <https://github.com/PanKbase/PanKbase-scRNA-seq>.

2. Cell annotations:

Cell type	Number of cells
Alpha	194443
Beta	120502
Acinar	50105
Ductal	33686
Active Stellate	16907
Delta	11680

Gamma + Epsilon	7087
MUC5B+ Ductal	3515
Endothelial	3385
Immune	2755
Cycling Alpha	2738
Quiescent Stellate	2132

Methods

1. Data collection:

1.1. Single-cell (sc) RNA-seq from the Human Pancreas Analysis Program (HPAP): scRNA-seq data and metadata generated from islet tissues using the 10X Chromium platform were downloaded from <https://hpap.pmacs.upenn.edu/> (accessed March 2024) [1], [2], [3], [4].

1.2. Single-cell (sc) RNA-seq generated using islets distributed by [Integrated Islet Distribution Program \(IIDP\)](#), and [Prodo Labs \(Prodo\)](#):

Literature searches were conducted in order to identify original studies involving scRNA-seq generated using islets distributed by IIDP and Prodo. Briefly, pancreatic islet datasets of expression profiling by high throughput sequencing were identified via GEO search for the date range before July 2024, then studies with scRNA-seq data were determined manually. Datasets were then downloaded from the studies' corresponding GEO page using the *fastq-dump* command from the SRA toolkit (v3.1.1) [5]. Data from the following GEO accession IDs were used: GSE142465 [6], GSE159556 [7], GSE183568 [8], GSE190447 [9], GSE201256 [10], GSE217837 [11], GSE221156 [12], GSE251730 [13], GSE251912 [14].

2. Data preprocessing:

The quality of raw reads was assessed using FastQC (v0.11.9) [15], then individual samples' reports were compiled together using MultiQC (v1.11) [16]. These reports were inspected in order to detect technical issues such as flow cell problems, base calling errors, and the presence of unexpected technical sequences. Based on the Per Tile Quality and Quality Score plots, we subsequently included sequencing runs with more than 50% of tiles indicated as high quality (i.e., the quality score was at or above the average) across all bases.

STARsolo (v2.7.10a) [17] was used to align raw reads to the human genome build hg38. Comprehensive annotation from Gencode release 39

(https://www.gencodegenes.org/human/release_39.html) was used to build STAR index files with the command *STAR --runMode genomeGenerate* and default parameters. As a

result, count matrices quantifying various features such as gene counts were obtained. For downstream analyses, count matrices GeneFull_ExonOverIntron were used. Additionally, MultiQC was used to compile log files output by STARsolo, and the report was used to examine various relevant metrics such as percent of uniquely mapped reads and mapping rates.

High quality reads were obtained using SAMtools (v1.14) [18] and the following command: *samtools view -h -b -q 255 -F 4 -F 256 -F 2048 \$bam*.

3. Quality control:

- In order to confirm donor metadata and check for sample swaps, we compared the read profiles with genotyping data using the mbv tool [19] included in QTLtools (v1.3.1) [20]. When donor unique identifiers were not supplied in the original studies, we assigned donor identities to samples when “perc_het_consistent” and “perc_hom_consistent” from mbv are both larger than 0.8. When donors did not have genotype data, we manually searched for individuals with matching age, sex, BMI and allocation centers to assign donor identifiers to samples. Subsequently, we retained only samples with a determined identifier.
- To systematically inspect quality metrics underlying all libraries, we utilized the UMI numbers and their ranks using the following algorithm. We ranked every barcode b based on their UMI numbers (n_b) in a decreasing order, then considered $\log(n_b)$ as a function of the rank's logarithm. In typical single-cell libraries, we expect this function to have two “knee” points corresponding to the transition between three subsets of barcodes: those with large, intermediate, and small n_b [21]. These barcode subsets generally represent cell-containing droplets, a mix of empty and non-empty droplets, and the barcodes associated with empty Gel Beads-in-emulsion, respectively [21], [22]. Next, we calculated the first-order discrete difference between two consecutive barcodes to obtain the rate of changes in n_b along the rank axis, then smoothed out this difference signal profile using the Savitzky–Golay filter [23]. As a result, we identified points along the profile with the largest change rates, which aligned with the “knees”, and quantified the “knee” point numbers. We retained all libraries with at least two “knee” points. For the libraries with one knee point, we manually inspected not only the barcode rank behaviors but also other quality control metrics (listed below) to decide if they would be retained.
- To distinguish potential cells from empty droplets, EmptyDrops [22] from the DropletUtils (v1.26.0) [24] was used. Droplets with significant deviations from the ambient profile at FDR < 5% were regarded as non-empty.
 - Adopting the mathematical models underlying EmptyDrops, we detected the end of the cliff point from the barcode rank plots of each sample. The cliff point is defined as the point with log-rank higher than that of inflection points (detected using EmptyDrops) and minimizes the signed curvatures.
- Ambient RNA was corrected using CellBender (v0.3.0) [25] in two rounds.

- First round (default run): Used a false positive rate (FPR) of 0.05 with all other parameters set to default.
- Second round (modified run): Set parameters as follows:
--expected_cells: number of barcodes identified as corresponding to cells
and --total_droplets_included: average of UMI numbers between inflection and end of cliff point.
- In order to identify barcodes that correspond to cells, we defined the following criteria:
 - Probability that a barcode corresponds to a true cell (obtained from CellBender using default settings) ≥ 0.99 .
 - Cells were detected as non-empty using EmptyDrops.
 - Cells with fractions of ambient reads $<$ a dynamic threshold determined per sample using the Multi-Otsu Thresholding algorithm on non-empty cells detected using EmptyDrops.
 - Cells with percent of chrMT reads $<$ a dynamic threshold determined per sample using the Multi-Otsu Thresholding algorithm on the combination of two sets: non-empty cells, and cells whose chrMT read fraction < 0.3 and rank higher than that of cliff end points.
- We retained samples with a number of qualified cells > 200 for downstream analyses.
- a. Processing of HTO samples:
 - HTO matrix files and barcode-treatment maps were downloaded from GEO page (<https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE251912>)
 - HTODemux() function from the Seurat package (v4.4.0) [26] was used to demultiplex the cells to their original treatments.

2. Doublet detection:

For each sample, DoubletFinder (v2.0.3) [27] was used twice on the RNA ambient-corrected gene count matrix obtained using modified runs of CellBender, and the second round of DoubletFinder was run without doublets detected in the first round. Next, we jointly clustered libraries across each tissue source (HPAP, IIDP and Prodo) using Seurat with doublets included, using Harmony (v1.2.0) [28] to correct for the following covariates: sex, BMI, age, studies, treatments and chemistry, and resolution of 1.8. We removed doublets and doublet-enriched clusters which are defined as ones with doublet rates $> 65\%$.

3. Data integration:

All data integration was performed using Seurat. In particular, due to the large number of samples, we first combined all samples per source (data generated using HPAP, IIDP, and Prodo samples) to create three Seurat objects, retaining cells that expressed at least one protein-coding gene, and protein-coding genes that are expressed in at least one cell. Subsequently, we merged all objects and carried out integration using Harmony, correcting for the following covariates: sex, BMI, age, studies, treatments,

chemistry and tissue sources. As a result, we obtained a set of 481,154 cells expressing 19,546 protein-coding genes.

Next, we adopted an iterative approach to further clean up the data.

- Step 1: We clustered the cells (resolution = 1.8), and tested for clusters with significantly different profiles of gene numbers and UMI numbers using Wilcoxon rank sum test. Here, we define a fold change as $\text{median}(\text{tested cluster's metrics}) / \text{median}(\text{all other clusters' metrics})$. A cluster is determined to be significantly different if their adjusted p-value < 0.05 and fold change > 2 , and they are removed from the subsequent analysis. As a result, we obtained a set of 476,429 cells, expressing 19,546 protein-coding genes.
- Step 2: We focused on identifying doublet-like cells using the remaining cells. For each sample, we began by clustering cells at a resolution of 4. We pinpointed clusters that were enriched with cells exhibiting high UMI counts and expressing markers from at least two cell populations in 50% or more of the cells. To achieve this, we performed Binomial tests to assess whether a cluster had more cells with UMI counts exceeding the 90th percentile of the samples than the expected rate of 0.1, using a Benjamini-Hochberg adjusted P-value threshold of < 0.05 . Cells within these enriched clusters were labeled as "doublet-like cells." Next, we employed Fisher's exact test to identify clusters in the integrated map that were enriched with doublet-like cells (Benjamini-Hochberg adjusted P-value threshold of < 0.05). Clusters in the integrated map that both were enriched with doublet-like cells and expressed markers from at least two cell populations in 50% or more of the cells were subsequently removed. Additionally, all doublet-like cells were removed.
- Step 3: We re-clustered and harmonized the data (resolution = 1.8), resulting in 48 clusters. Using the following list of markers, we annotated the cell clusters to 13 major cell types:
 - Beta: 'INS', 'IAPP'
 - Alpha: 'GCG'
 - Delta: 'SST'
 - Gamma: 'PPY'
 - Epsilon: 'GHRL'
 - Ductal: 'CFTR', 'KRT19'
 - Acinar: 'REG1A', 'CTRB2', 'PRSS1', 'PRSS2', 'CPA1'
 - Active Stellate: 'PDGFRB', 'COL6A1'
 - Quiescent Stellate: 'PDGFRB', 'COL6A1', 'RGS5'
 - Endothelial: 'PECAM1', 'PLVAP', 'ESAM', 'VWF'
 - Immune: 'PTPRC'
 - Cycling Alpha: 'GCG', 'MKI67', 'CDK1'

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Citation

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R Object's Metadata README

1. 'cell_type': Cell annotations.
2. 'ncount_rna': Number of UMIs for barcodes that expressed at least one non-duplicate protein-coding gene
3. 'nfeature_rna': Number of non-duplicate protein-coding genes that are expressed in at least one barcode
4. 'rna_total_reads': Total number of reads aligned to **all genes** (not just non-duplicate protein-coding genes) obtained using STARsolo GeneFull_ExonOverIntron setting.
5. 'rna_uniquely_mapped_reads': Number of uniquely aligned reads to **all genes** (not just non-duplicate protein-coding genes) obtained using STARsolo GeneFull_ExonOverIntron setting
6. 'rna_secondary_alignments': Number of supplementary aligned reads to **all genes** (not just non-duplicate protein-coding genes) obtained using STARsolo GeneFull_ExonOverIntron setting
7. 'rna_umis': Number of UMIs per barcode prior to CellBender correction, using **all genes** (not just non-duplicate protein-coding genes)
8. 'rna_pct_mitochondrial': Fraction of mitochondrial reads, calculated as number of reads aligned to chrMT divided by total number of uniquely aligned reads
9. 'samples': Sample names
10. 'barcodes': Original barcodes, without sample name appended
11. 'treatments': Treatments of samples. Samples that did not go through any treatments are “no_treatment”, otherwise, the treatments are noted according to individual studies. Samples from the study “GSE251912” were generated using HTO assays; therefore, cells from the same samples may have different treatments
12. 'chemistry': Chemistry of 10x kits used for data generation
13. 'source': Tissue source
14. 'center_donor_id': Donor center ID identifier for cross-referencing
15. 'rrid': Research Resource Identifier, with “RRID:” as prefix
16. 'donor_accession': Unique identifier (prefixed with PKB, assigned by server)
17. 'sex': Self-reported gender
18. 'age_(years)': Age of the donor in years (numeric value).
19. 'aliases': Lab-specific identifiers
20. 'bmi': Body mass index in kg/m²
21. 'c_peptide_(ng/ml)': C-Peptide concentration in ng/ml.
22. 'cause_of_death': Primary medical condition that led to death.
23. 'reported_ethnicity': Self-reported ethnicity of the donor
24. 'description_of_diabetes_status': Indicates the specific type of diabetes or absence of it. It provides more detail than just a yes/no classification.
25. 'diabetes_duration_(years)': Duration of diabetes in years.
26. 'diabetes_status': Ontology Term for diabetes status (MONDO IDs)
27. 'hba1c_(percentage)': HbA1C percentage measurement
28. 'family_history_of_diabetes': Boolean indicating family history of diabetes

29. 'family_history_of_diabetes_relationship': Array describing relationship to diabetic family members
30. 'glucose_lowering_therapy': Type of therapy for managing blood glucose levels
31. 'aab_gada_positive': Boolean indicating presence of GADA autoantibodies
32. 'aab_gada_value_(unit/ml)': Numeric value of GADA autoantibodies in unit/ml
33. 'aab_ia2_positive': Boolean indicating presence of IA2 autoantibodies
34. 'aab_ia2_value_(unit/ml)': Numeric value of IA2 autoantibodies in unit/ml
35. 'aab_znt8_positive': Boolean indicating presence of ZNT8 autoantibodies
36. 'aab_znt8_value_(unit/ml)': Numeric value of ZNT8 autoantibodies in unit/ml
37. 'aab_iaa_positive': Boolean indicating presence of IAA autoantibodies
38. 'aab_iaa_value_(unit/ml)': Numeric value of IAA autoantibodies in unit/ml
39. 'biosample_accession': Unique identifier for biosamples (prefixed with PKB, assigned by server)
40. 'biosample_cold_ischaemia_time_(hours)': Duration in hours that the pancreas was kept at a low temperature after removal from the donor
41. 'biosample_hand_picked': Whether the pancreas or its components were manually selected or processed
42. 'islet_function_available': Whether functional assays or data are available for the isolated pancreatic islets pre- or post-shipping (boolean)
43. 'ieq_pancreas_weight_(grams)': Ratio of Islet Equivalents to the weight of the pancreas
44. 'islet_yield_ieq': Total number of Islet Equivalents obtained from the pancreas
45. 'organ_source': Type of organ donor from which the pancreas or pancreatic tissue was obtained (deceased, living, other classifications, unknown)
46. 'isolation_center': Facility or location where the islets were isolated, including islet isolation centers affiliated with the Integrated Islet Distribution Program (IIDP), IsletCore at the University of Alberta (Edmonton), Human Pancreas Analysis Program (HPAP), and Prodo.
47. 'biosample_warm_ischaemia_duration_down_time_(hours)': Duration in hours that the pancreas was without blood supply at body temperature
48. 'hospital_stay_(hours)': Total hours of hospitalization
49. 'donation_type': Type of organ donation (Donation after Circulatory Death, Donation after Brain Death, Natural Death Donation, Medical Assistance in Dying)